just `0` or `1`, they can intersect multiple times: one more reason to prefer other styles over polylines.

15.4.3 Bézier Curves

An interesting and flexible alternative to polylines is given by Bézier curves; these curves are a valuable solution, because they can be drawn with a variable degree of precision, depending on the computational resources available: Bézier curves have a beautiful mathematical formula, and are both flexible and precise. Going into the details of how they work is beyond the scope of this book, but we'll try to give a quick introduction that should be enough to get you started. To the interested reader that would like to delve into this topic, we have two freely available online resources:

1. Sederberg, Thomas W. "Computer aided geometric design" (2012).
2. "A Primer on Bézier Curves", https://pomax.github.io/bezierinfo/.

Let's start building our intuition with the geometric definition of Bézier curves, shown in figure 15.21 for a quadratic curve.

A quadratic curve requires three points: two endpoints (`S` and `E` in figure 15.21) and a control point (C_1); first we have to draw the two segments between the endpoints and C_1; then, we need to choose two points on those segments, under the constraint that if the point on SC_1 (call it `A`) is such that $SA/SC=t$, then `B`, the endpoint on EC_1, must satisfy $EB/EC_1=(1-t)$.